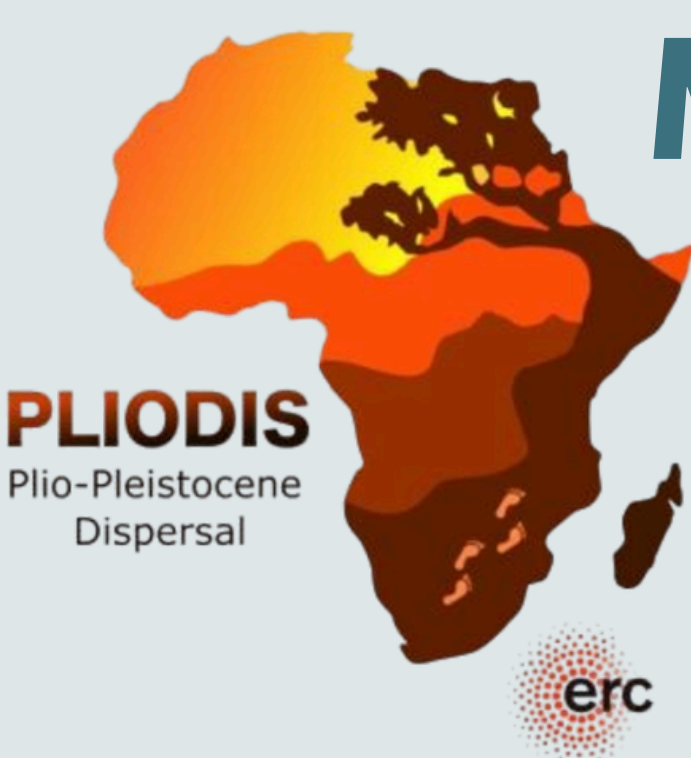




Multi-trophic patterns in mammalian joint and bone morphology and their potential for palaeoenvironmental reconstruction

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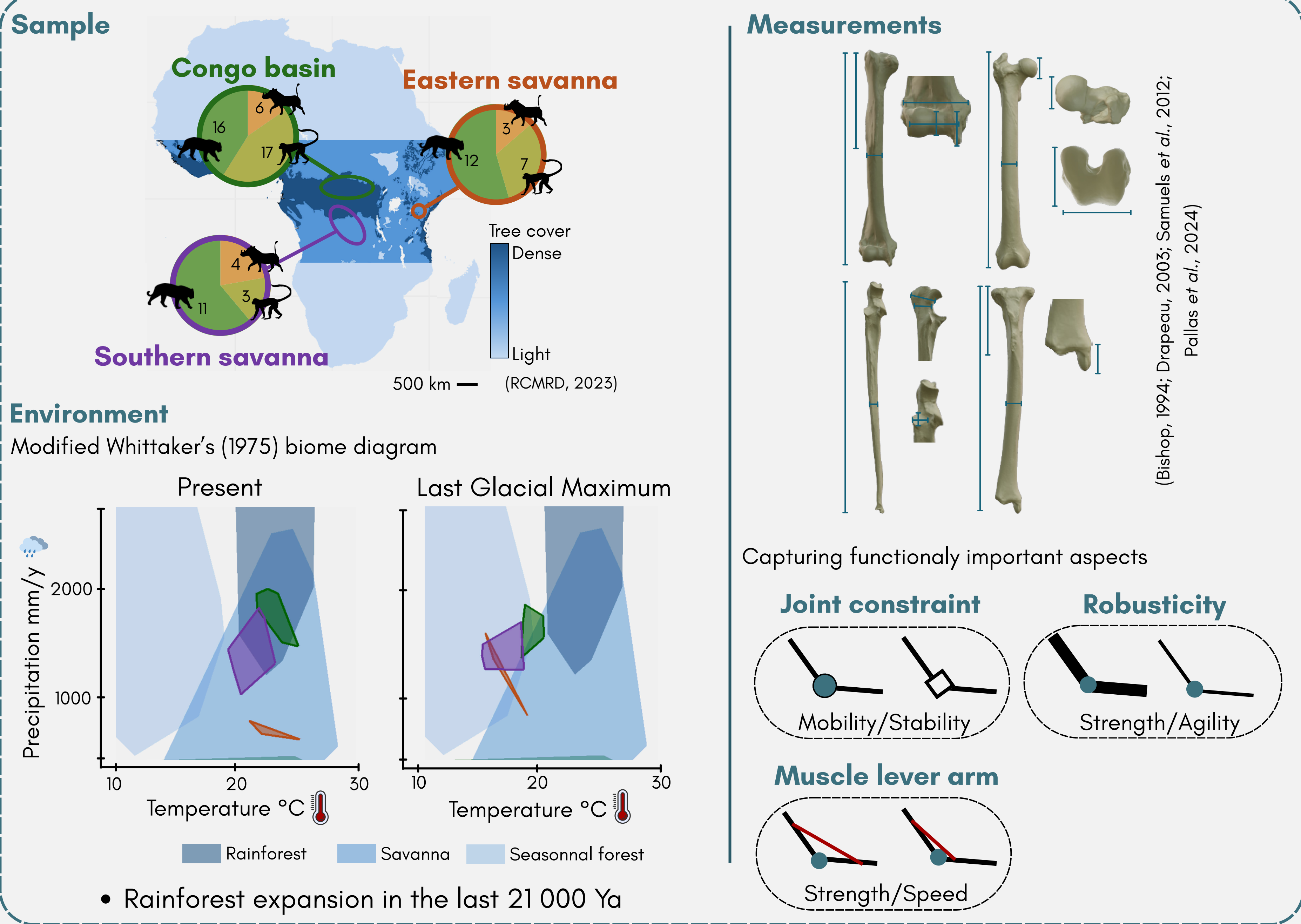
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Questions

- Is **ecomorphology** a **reliable tool** to reconstruct past environments?
- Do **clades** respond similarly to environmental/habitat changes?

Material and context



Method

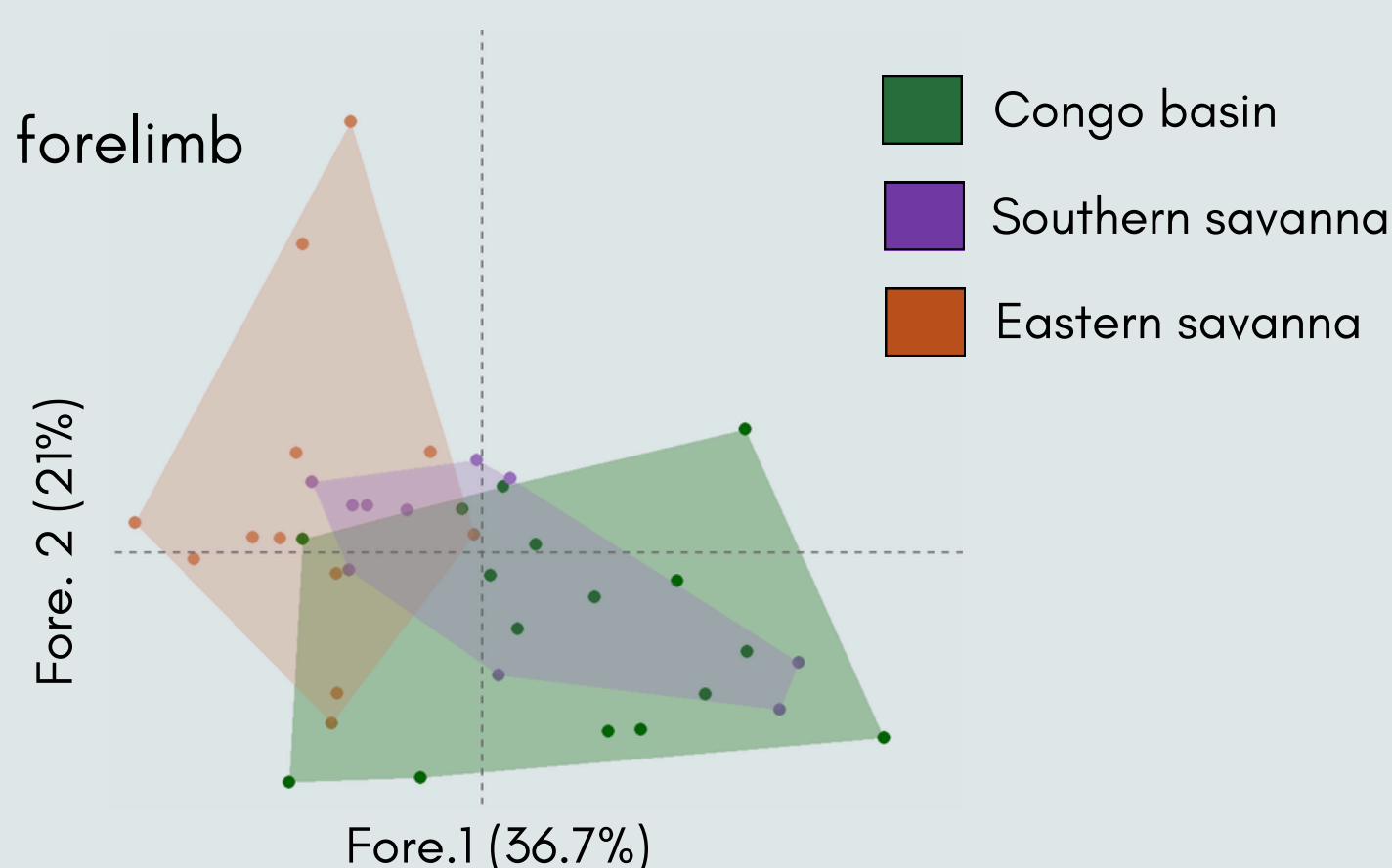
Workflow

1) PCA for each of the **limbs**, by **Taxa**

- Summary and description of the morphologies with PC dimensions

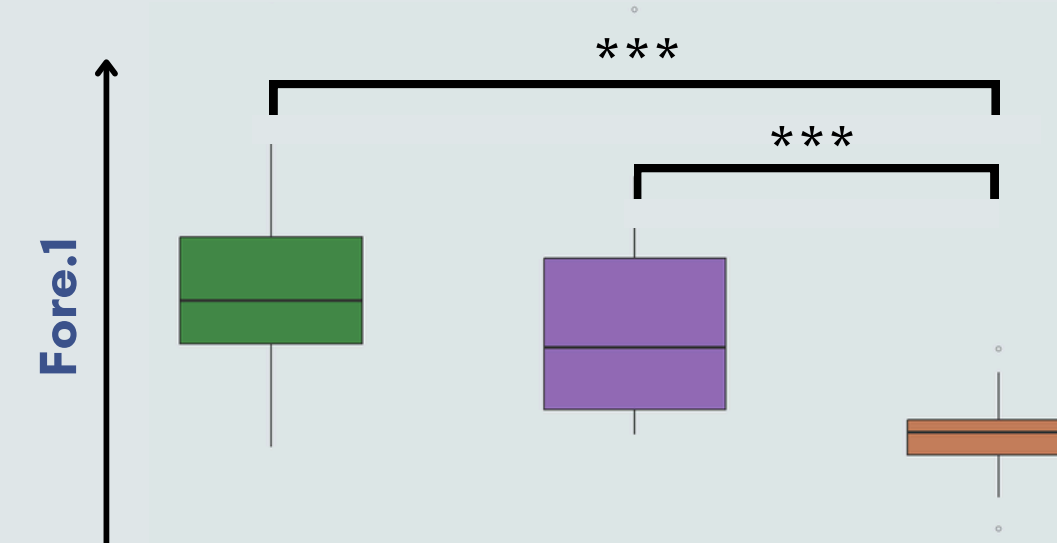
Example:

Carnivore's forelimb



2) **Comparison** of the PC dimensions between **habitats**

- Wilcoxon Tests



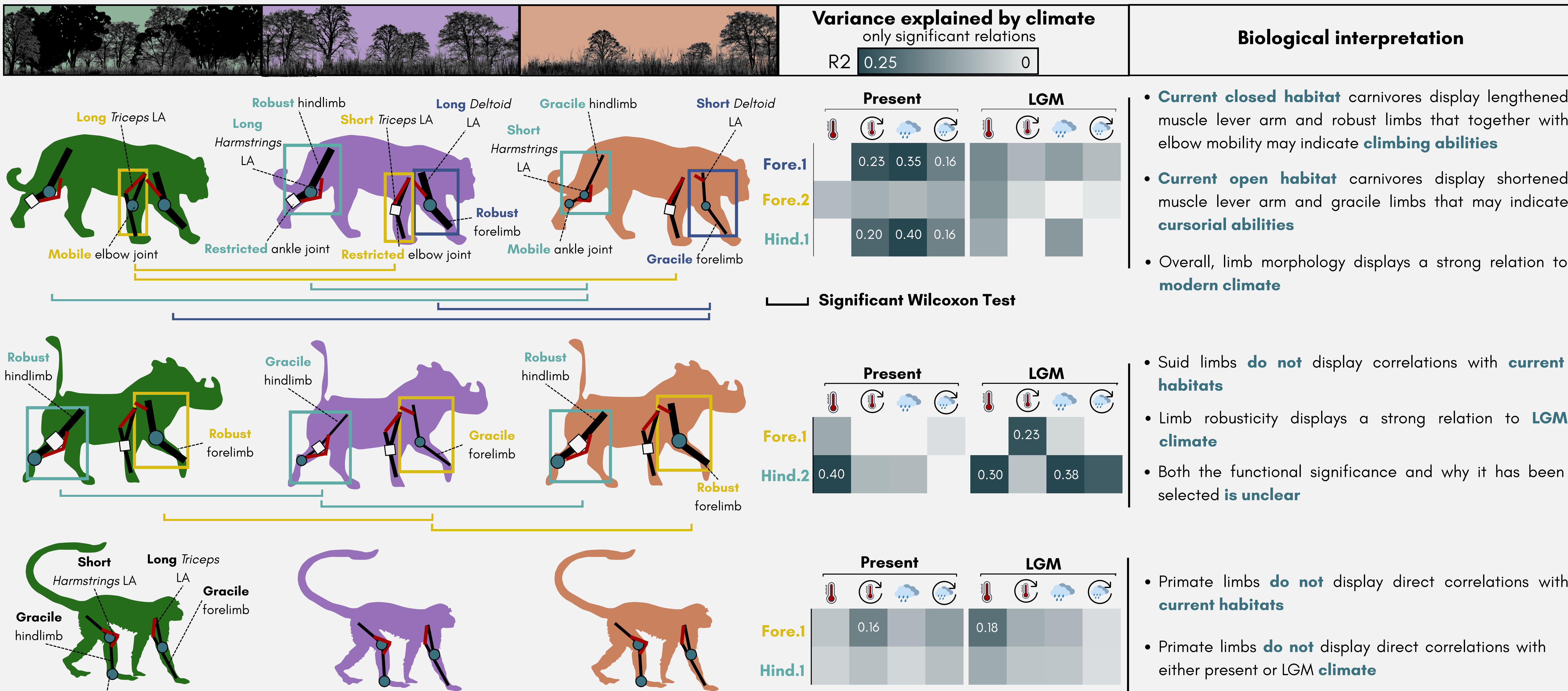
3) Testing the **relationship** between PC dimensions and **climate**

- Linear regressions between PC dimensions and present and LGM climatic variables

- Temperature
 - Temperature seasonality
 - Precipitation
 - Precipitation seasonality
- (Beyer *et al.* 2020)



Results



Take home message & future work

- Carnivores display specific adaptations to **current habitats**, thus, making them useful clades for reconstructing palaeoenvironments
- Suids and primates **do not** display specific morphological adaptations to current habitats or relation with present climate
- Primate **morphological diversity** is a useful model to reconstruct substrate diversity, which relates to **current** composition of the **tree coverage**
- Multi-trophic analyses considering **prey and predator dynamics** could bring further palaeoenvironmental reconstructions by cross validating **environmental filtering** signals

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